

Suvash Shah Thakuri

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PROFESSIONAL SUMMARY

Information Technology Specialist I with the California State Controller's Office and a software engineer with 15+ years of experience designing, delivering, and modernizing enterprise systems for state government, automotive manufacturing, and biotech research. Proven track record of leading feasibility studies, conducting systems analysis, recommending technology solutions, and guiding complex IT projects from concept to production. Currently a PhD student in Computer Science at North Dakota State University with a research focus on software security and systems modernization. Seeking to advance into the Information Technology Specialist II classification to lead larger, cross-program technology initiatives at the State of California.

ALIGNMENT WITH IT SPECIALIST II DOMAINS

- **Software Engineering:** Architecture and full-stack development on .NET Core, C#, SQL Server, Web API, and Blazor across government, enterprise, and research environments.
- **Information Security Engineering:** Secure application development, access control design, API security, and applied research in software and systems security at the PhD level.
- **System Engineering:** Feasibility studies, system design, integration, and modernization of legacy state and enterprise platforms.
- **IT Project Management:** Leading milestones, deliverables, cross-team coordination, and stakeholder communication on multi-program state initiatives.

PROFESSIONAL EXPERIENCE

Information Technology Specialist I

Aug 2024 – Present

California State Controller's Office (SCO) – Sacramento, CA

Program Support & Technical Leadership

- Support the School Facility Program (SFP) and lead development of the Transportation Development Act (TDA) program, coordinating delivery with audit users, business analysts, and project management.
- Lead modernization of the Audit Management System version 2 (AMS2), including migration from .NET Core 2.0 to .NET 8.0, secure application design, and integration with downstream reporting.
- Designed and implemented a data synchronization pipeline from the LEA project into AMS2 for Entity and CPA data, eliminating manual reconciliation between systems.
- Refactor existing .NET code and SQL Server schema to remove technical debt without impacting the development schedule or delivery timeline.
- Design SSRS reports and secure backend services used by multiple State Departments; recommend system upgrades and process improvements based on root-cause analysis of operational issues.
- Automated the end-to-end OPSC file import process for six State of California programs, reducing manual processing time and improving data integrity across financial workflows.

Senior Software Engineer

2021 – 2024

Tesla, Inc. – Fremont, CA

- Engineered modules of *Warp*, Tesla's in-house ERP used to plan, procure, and produce vehicles, on a stack of .NET Core 6, C#, Angular 12, Web API, and SQL Server.
- Built a global-scale Material Requirements Planning (MRP) system handling inventory, procurement, and forecasting across 1,200+ warehouses worldwide.
- Implemented optimized SQL solutions and API security protocols; conducted root-cause analysis on critical incidents requiring immediate action.
- Partnered with product owners, QA, and release engineers to own features from design through deployment, including unit tests, integration tests, and regression coordination.

Technical Staff

2020 – Present

Dhuni Software LLC – California

- Lead design and delivery of applications on .NET Core, SQL Server, MongoDB, Elasticsearch, and D3.js for clients in biotech research and scientific computing.
- Designed a graph-based visualization tool representing stages of a large-molecule purification process, enabling research scientists to explore experimental data interactively.
- Built an integrated review and approval platform for Assay Requests, replacing a manual paper-based workflow with a standardized digital process.

.NET Full Stack Developer

2018 – 2020

County of San Mateo – Redwood City, CA

- Modernized legacy financial systems used by County agencies, adding access control, budget monitoring, and real-time reporting capabilities.
- Tuned Oracle queries and ADO.NET data access layers, materially improving performance for agency-facing reports.
- Built a .NET / SQL role-based access control tool used across multiple County applications.
- Conducted stakeholder interviews to identify pain points and translate institutional requirements into technical solutions.

Software Engineer / Application Developer

2010 – 2017

Deerwalk Nepal · Bracket Global LLC · 12th Wonder LLC

- Developed browser automation tools, healthcare analytics, and scheduling software on .NET Framework, SQL Server, and Crystal Reports.
- Led early development of biometric attendance systems, including fingerprint recognition integration with enterprise applications.

EDUCATION

Ph.D., Computer Science (Software and Security)

2025 – Present

North Dakota State University – Fargo, ND

Advisor: Dr. Malik. Research direction: systems modernization and software security (C-to-Rust migration and related areas).

M.S., Computer Science

2017

Maharishi International University – Fairfield, IA

B.E., Computer Engineering

2010

Institute of Engineering, Tribhuvan University – Nepal

TECHNICAL SKILLS

- **Languages:** C#, SQL, JavaScript / jQuery
- **Frameworks & Web:** .NET, ASP.NET MVC, Web API, Blazor, Bootstrap, HTML, CSS
- **Databases:** Microsoft SQL Server, Oracle, MySQL, Firebase, MongoDB
- **Tools & Practices:** Visual Studio, VS Code, Git, SVN, Jira, SSRS, Agile, TDD, SOLID, Design Patterns

ACADEMIC SERVICE & COMMUNITY LEADERSHIP

- **Commissioner**, Community Advisory Commission, City of Pittsburg, California – advise city leadership on community concerns, equity, and development initiatives.
- **Board of Directors**, Nepali Association of Northern California – strategic planning, community programs, and advocacy.
- **Founder**, Marpha – nonprofit building open-source technology solutions for rural Nepal.
- **Instructor (2012–2014)**, Deerwalk Institute of Technology, Tribhuvan University – undergraduate .NET programming and application design.

CERTIFICATIONS & MEMBERSHIPS

- Microsoft Certified Professional
- Microsoft Specialist in C# Programming
- Member, IEEE Computer Society